

ASTERIA

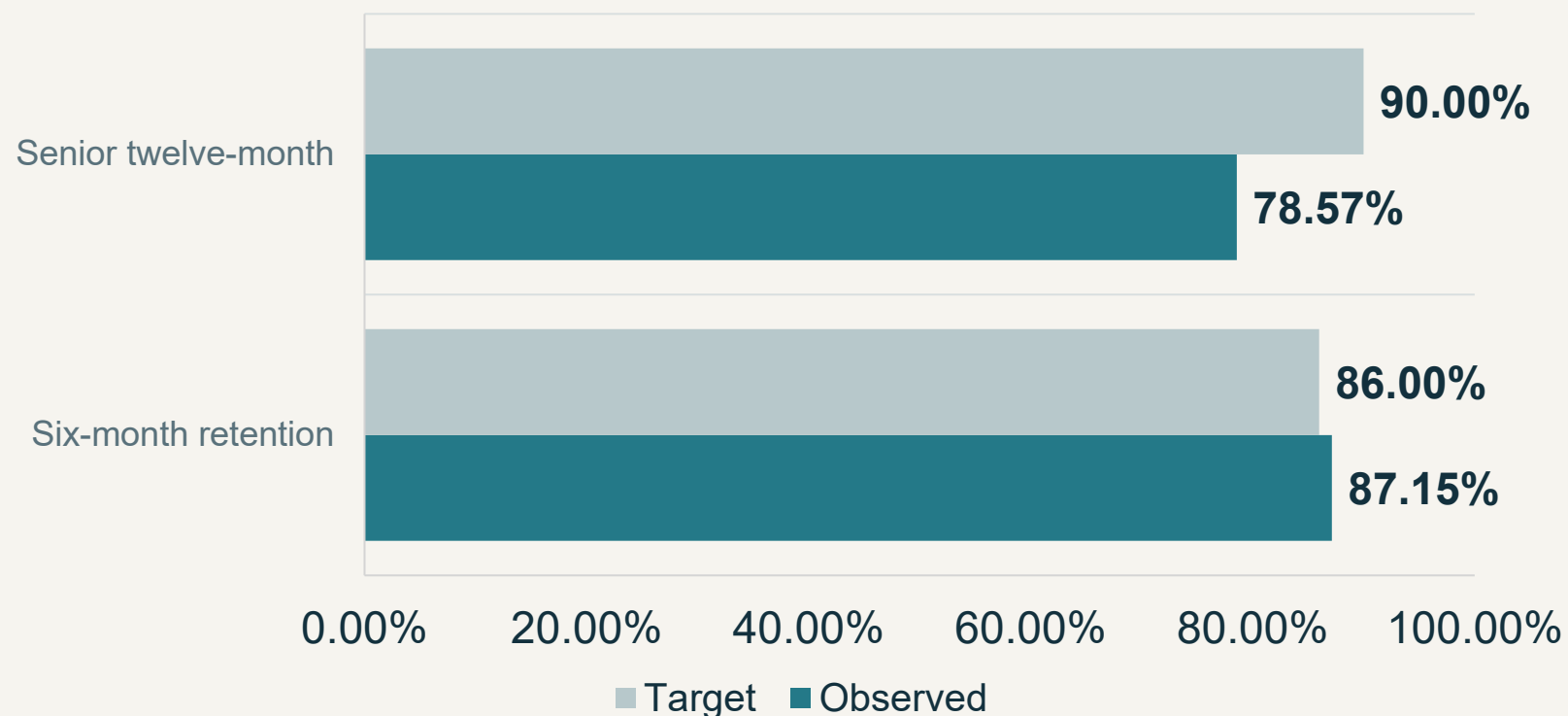
# Workforce retention

Results and evidence

31 December 2025 snapshot  
Synthetic workforce data



# Senior retention falls below target



**11.43**

percentage points  
below the senior  
target

Regretted turnover: 5.10% against a 7.50% ceiling

# Cleaning preserves uncertainty

**2,407**

supplied rows

**2,400**

unique records after 7 exact duplicates

**8 country aliases standardized**

9 missing countries remain unknown

10 invalid-date records quarantined

Raw data stays unchanged. Every exclusion remains auditable.

# Retention starts with eligible cohorts

6 months

**1,580 / 1,813**

87.15% retained

12 months, senior

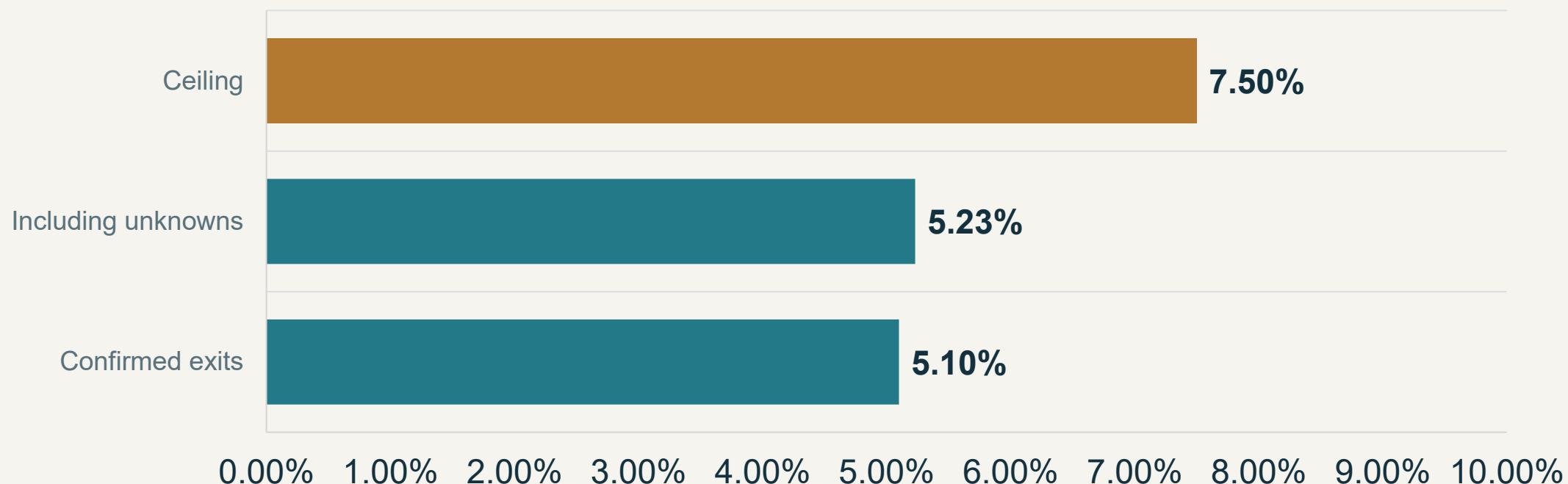
**209 / 266**

78.57% retained

Only hires who reach the milestone by the snapshot enter the denominator.

Hiring period: 2021-2025. Termination on the anniversary counts as retained.

# Turnover stays below its ceiling



**82 exits / 1,607.52 average daily headcount**

The upper scenario adds 2 potentially regretted exits. It is not a confidence interval.

# When a cohort is too recent

Senior retention, Hire year 2025

# Unavailable

0 mature hires

The dashboard keeps pending hires outside the denominator.  
Company-wide KPI cards remain fixed while cohort filters change.

# Historical context has a time boundary

## Published before the cutoff

Reference period ends by the cutoff

Maximum source age

**120 days**

Unemployment

**240 days**

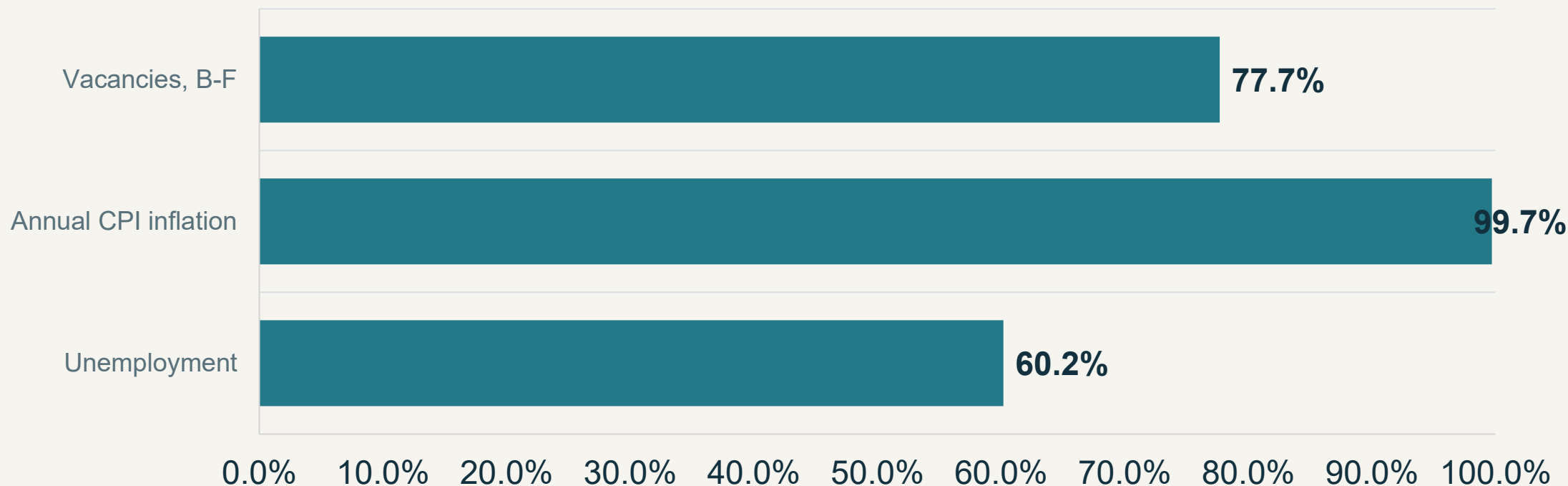
Vacancy rate

**730 days**

Annual CPI

Retention cutoff: hire date. Turnover cutoff: start of the annual window.

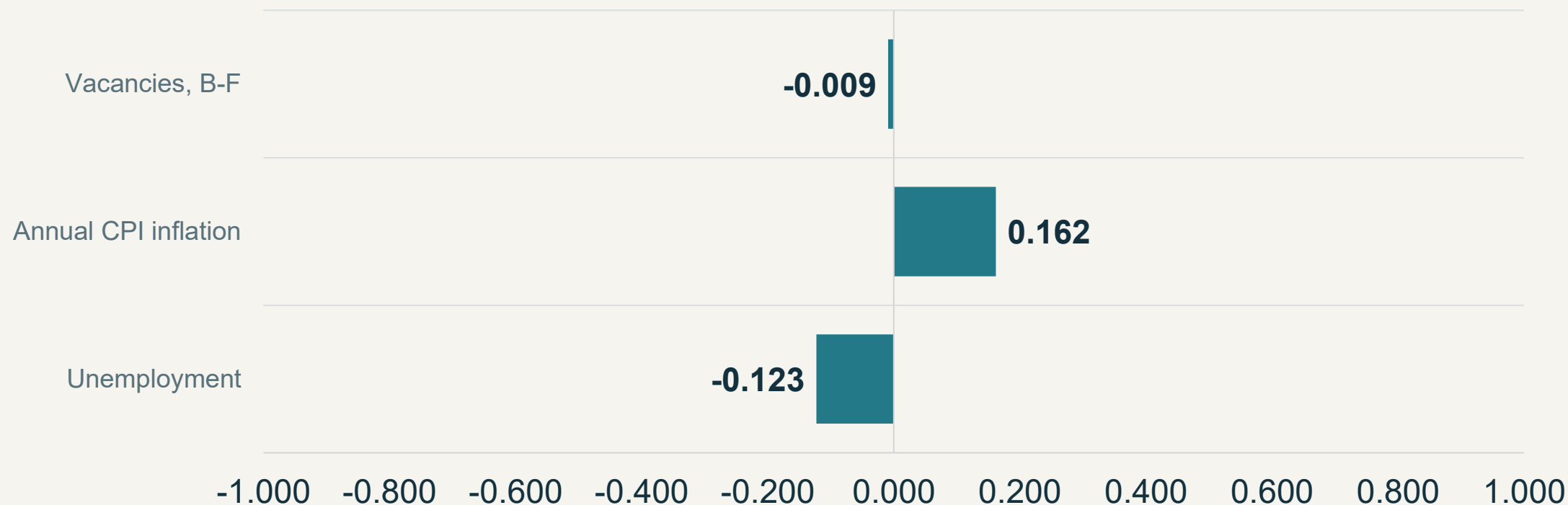
# Coverage changes with the indicator



**1,813 mature hires in the six-month population**

B-F covers industry and construction. Missing evidence does not remove KPI records.

# Weak associations limit interpretation



## Six-month retention in synthetic matched cohorts

Small cohorts, reused vintages and selection prevent causal conclusions.

# The working solution uses Python and HTML

## 01 Python

Cleaning, metrics and temporal joins

## 02 SQLite

Reconciled reporting views

## 03 HTML

Interactive dashboard and audit context

One entry point: `scripts/run_pipeline.py`. Saved evidence supports offline replay.

# Proposed production architecture

## Azure Data Factory

Schedules acquisition and processing

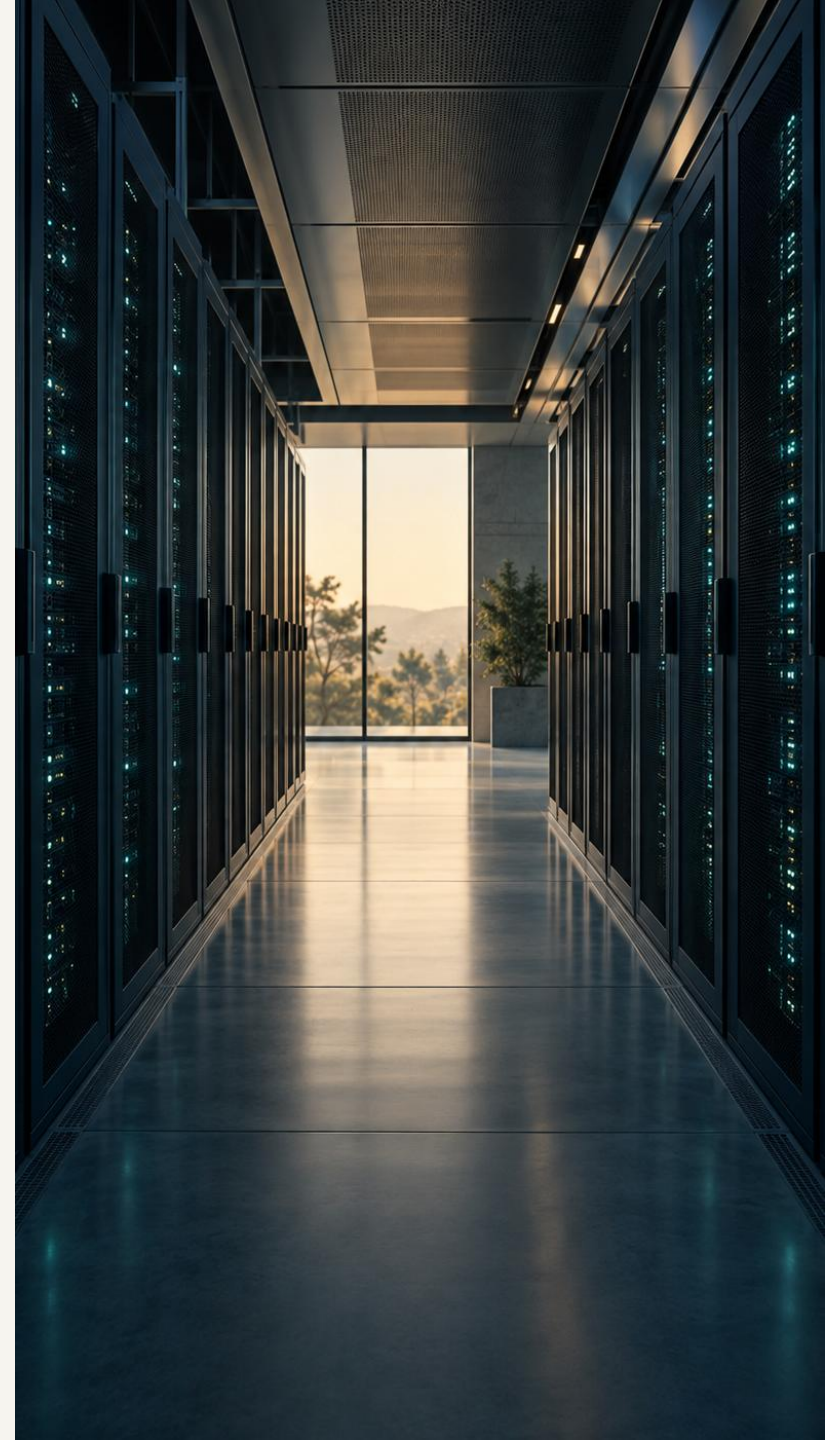
## ADLS and Databricks

Preserve evidence and build validated Delta tables

## Approved release, SQL and Power BI

Publish only a consistent, validated run

**Future design only. No cloud deployment.**



# What the evidence supports

## Senior retention is the priority

Investigate internal drivers of the shortfall

Resolve missing countries and unknown classifications

Expand historical evidence before stronger economic claims

Questions